

# 42.510 Statistical Learning for Data Science

Data censoring and survival analysis

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Week 11, Lecture 2

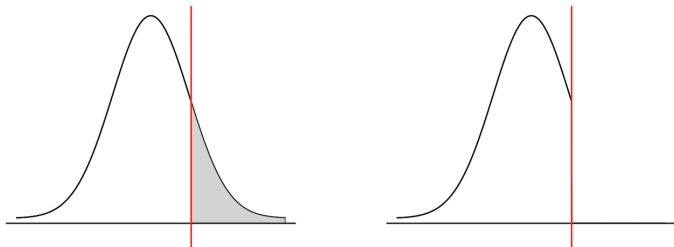
# Outline

- Data censoring
- Social science: extra-marital affairs
  - ▶ Tobit model
- Healthcare: survival analysis
  - ▶ Kaplan-Meier estimation
  - ▶ Cox proportional hazards model

# Where do we observe censored data?

- Survival of patients in a drug trial.
- Abandonment time (patience) of customers waiting in queue at some call center.
- Examinations/ tests.
- Limitation of the measuring instrument.

# Censored-dependent variable



- Suppose censored-dependent variable  $Y^* \sim \mathcal{N}(\mu, \sigma^2)$ .
- Consider censoring at  $C$  (say, capacity) above, so that we observe:

$$Y = \begin{cases} Y^*, & \text{if } Y^* \leq C, \\ C, & \text{otherwise.} \end{cases}$$

# Censored Tobit regression

- Proposed by James Tobin (1958). Based on the regression model:

$$\begin{aligned} Y_i^* &= \beta_0 + \sum_{j=1}^p \beta_j x_{ij} + \epsilon_i, \\ &= \mathbf{x}_i^\top \boldsymbol{\beta} + \epsilon_i, \forall i = 1, \dots, n. \end{aligned}$$

where  $\mathbf{E}[Y_i^* | \mathbf{X}_i = \mathbf{x}_i] = \mathbf{x}_i^\top \boldsymbol{\beta}$ .

- $\beta_1, \dots, \beta_p$  = model coefficients,
  - $x_{i1}, \dots, x_{ip}$  = predictor variables for the  $i$ -th individual,
  - $n = \#$  of observations, and,
  - $\epsilon_i \sim \mathcal{N}(0, \sigma^2)$  is the model error.
  - Let  $\phi, \Phi$  be the standard normal pdf and cdf.
- The observable variable  $Y_i$  is left-censored at 0:

$$Y_i = \begin{cases} Y_i^*, & \text{if } Y_i^* > 0, \\ 0, & \text{if } Y_i^* \leq 0. \end{cases}$$

# Censored Tobit regression

- In the Tobit regression model we want to

- ▶ estimate  $\beta$ ,
- ▶ find  $\mathbf{E}[Y_i | \mathbf{X}_i = \mathbf{x}_i]$ .

- First we find the distribution of  $Y_i$ . Clearly  $Y_i \geq 0$ .

$$\begin{aligned}\mathbb{P}(Y_i = 0 | \mathbf{X}_i = \mathbf{x}_i) &= \mathbb{P}(Y_i^* < 0 | \mathbf{X}_i = \mathbf{x}_i) = \mathbb{P}(\epsilon_i < -\mathbf{x}_i^\top \beta | \mathbf{X}_i = \mathbf{x}_i) \\ &= \mathbb{P}(\epsilon_i / \sigma < -\mathbf{x}_i^\top \beta / \sigma | \mathbf{X}_i = \mathbf{x}_i) = \Phi(-\mathbf{x}_i^\top \beta / \sigma) \\ &= 1 - \Phi(\mathbf{x}_i^\top \beta / \sigma),\end{aligned}$$

$$f_{Y_i | \mathbf{X}_i}(y_i | \mathbf{x}_i) = \frac{1}{\sigma \sqrt{2\pi}} e^{-\frac{(y_i - \mathbf{x}_i^\top \beta)^2}{2\sigma^2}} = \frac{1}{\sigma} \phi\left(\frac{y_i - \mathbf{x}_i^\top \beta}{\sigma}\right), \quad y_i > 0.$$

- Therefore log-likelihood at  $(\mathbf{x}_i, y_i)$  is

$$\ell_i(\beta, \sigma) = \mathbf{1}(Y_i = 0) \log \left[ 1 - \Phi\left(\frac{\mathbf{x}_i^\top \beta}{\sigma}\right) \right] + \mathbf{1}(Y_i > 0) \log \left[ \frac{1}{\sigma} \phi\left(\frac{y_i - \mathbf{x}_i^\top \beta}{\sigma}\right) \right].$$

- Find log-likelihood for a sample of size  $n$ . Find MLE (the function is concave).

# Interpreting the model output

- Now we find  $\mathbb{E}[Y_i | \mathbf{X}_i = \mathbf{x}_i]$ .
- We can compute:

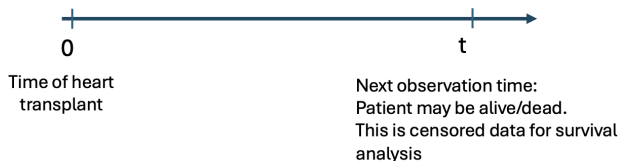
$$\mathbb{E}[Y_i | \mathbf{X}_i = \mathbf{x}_i] = \Phi\left(\frac{\mathbf{x}_i^\top \boldsymbol{\beta}}{\sigma}\right) \mathbf{x}_i^\top \boldsymbol{\beta} + \sigma \phi\left(\frac{\mathbf{x}_i^\top \boldsymbol{\beta}}{\sigma}\right).$$

- This is a non-linear function of  $\boldsymbol{\beta}$  unlike  $\mathbb{E}[Y_i^* | \mathbf{x}_i] = \mathbf{x}_i^\top \boldsymbol{\beta}$ .

# R implementation on extramarital affairs data

- Data: `extramarital.csv`
  - Use package: `survival`
  - Function call: `survreg(Surv(time,event,type="left")~.,data= ... ,dist="gaussian")`
- 
- *We will convert to a classification problem for prediction*

# Survival analysis in healthcare



- Suppose  $T$  = waiting time until a specific event happens (e.g., *death*).
- Let  $T$  have c.d.f  $F$  and p.d.f.  $f(t)$ .
- The *survival* function at time  $t > 0$  is given by

$$S(t) = \mathbb{P}(T > t) = 1 - F(t).$$

# Survival function and hazard rate

- Hazard function  $\lambda(t)$  or hazard rate gives the instantaneous survival rate/rate of failure.

$$\begin{aligned}\lambda(t) &= \lim_{\Delta t \rightarrow 0} \frac{\mathbb{P}(t < T \leq t + \Delta t | T > t)}{\Delta t} \\ &= \lim_{\Delta t \rightarrow 0} \frac{\mathbb{P}(t < T \leq t + \Delta t)}{\mathbb{P}(T > t)\Delta t} = \frac{f(t)}{S(t)}.\end{aligned}$$

where  $T$  has pdf  $f$ .

- Note  $\lambda(t) = \frac{d}{dt}(-\ln S(t))$
- We want to estimate  $S(t)$  or  $\lambda(t)$  from data (which may be censored).
- **Example:** Suppose  $T \sim \text{Exp}(\lambda)$  where  $\lambda > 0$ .
  - ▶ Then  $S(t) = \mathbb{P}(T > t) = e^{-\lambda t}$ .
  - ▶ Moreover  $\lambda(t) = \lambda$ , a constant.

# Kaplan-Meier estimator for the survival function

- Kaplan and Meier (1958) proposed a (non-parametric) estimator for  $S(t)$ .
- Denote by

$$t_1 \leq t_2 \leq \dots \leq t_k$$

the distinct ordered times of death (here  $k \leq N$  may not be  $N$ ).

- $d_i$  = number of deaths at time  $t_i$ .
- $n_i$  = number alive (*and under consideration, i.e. not censored*) just before time  $t_i$ . This is the number exposed to risk at time  $t_i$ .
- The Kaplan-Meier estimate of the survival function and hazard function are:

$$\hat{S}(t) = \prod_{i: t_i \leq t} \left(1 - \frac{d_i}{n_i}\right),$$

$$\hat{\lambda}(t) = \frac{d_i}{n_i}, \quad \text{for } t_i \leq t < t_{i+1}.$$

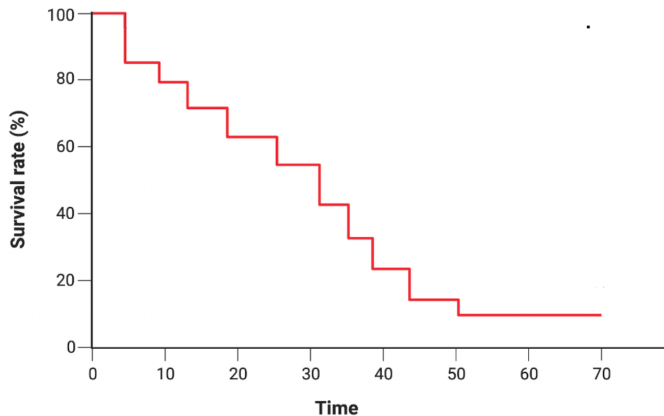
- With censoring,  $n_i$  is the number of people at risk minus the number of losses.

# Kaplan-Meier estimator for the survival function

- Consider the following remission times (in weeks) for 21 patients suffering from leukaemia:
  - ▶ 1,1,2,2,3,4,4,5,5,8,8,8,8,11,11,12,12,15,17,22,23.
- Survival function estimate

| $t$            | $\hat{S}(t)$                                 |
|----------------|----------------------------------------------|
| $t < 1$        | $\frac{21}{21} = 1$                          |
| $1 \leq t < 2$ | $1 \cdot \frac{21-2}{21} = 0.905$            |
| $2 \leq t < 3$ | $1 \cdot 0.905 \cdot \frac{19-2}{19} = 0.81$ |
| ...            | ...                                          |

# Kaplan-Meier curve



# Cox proportional hazards model

- Proposed by David Cox (1972).
- Used in bio-medical research.
- Adopted in Framingham heart study.
- Estimates hazard rate and hence survival rate (at an individual level).
- Describe hazard rate  $\lambda(t)$  as

$$\lambda(t) = \lambda_0(t)e^{\beta_1 x_{i1} + \beta_2 x_{i2} + \dots + \beta_p x_{ip}} = \lambda_0(t)e^{\beta^\top \mathbf{x}_i}.$$

- $\beta = (\beta_1, \dots, \beta_p)^\top$  = model coefficients.
- $\mathbf{x} = (x_{i1}, \dots, x_{ip})^\top$  = predictor variables for  $i$ -th individual.
- *baseline hazard rate*  $\lambda_0(t)$ : hazard where all predictors are 0.

# Cox proportional hazards model

$$\lambda(t) = \lambda_0(t)e^{\beta_1 x_1 + \beta_2 x_2 + \dots + \beta_p x_p} = \lambda_0(t)e^{\beta^\top \mathbf{x}}.$$

- $\beta_j > 0$  means as  $x_{ij} \uparrow$ , we have  $\lambda(t) \uparrow$ . So length of survival  $\downarrow$ .
- $t_i$  = time when individual  $i$  exists, likelihood of this is:

$$\ell_i(\beta) = \frac{e^{\beta^\top \mathbf{x}_i}}{\sum_{j \in R(t_i)} e^{\beta^\top \mathbf{x}_j}},$$

- where  $R(t_i)$  are all individuals at risk at time  $t_i$ .
- $\beta$  can be estimated by solving an MLE problem. The likelihood function is concave and hence a solution can be obtained numerically.
- This can be extended to censored loss.

# R implementation on heart transplant data

- Data: `heart.csv`
- Use package: `survival`
- Function call: `survfit(Surv(start,stop,event)~1,data=...)`

# References

- Teaching notes
- James et al. (2023), *Introduction to Statistical Learning with R*, Chapter 11.